



PROJECT BRIEF: MOSSBACK ARCADE - AN ONLINE GAMING WEBSITE

1. Company Details

Field	Detail
Company Name	Mossback Games
Company Type	Independent studio
Industry	Game Development, Casual Browser Games
Product	Mossback Arcade (online gaming website)
Category	Web Development
Technology Stack	HTML5, CSS3, Vanilla JavaScript (no frameworks, no backend)

2. Project Title

Design and development of Mossback Arcade, a responsive online gaming website where visitors browse a catalog of casual games and play them in the browser. The developer builds five games for the launch, and the catalog supports adding more games over time so the studio can keep growing it from its own side. The site is built with HTML5, CSS3, and vanilla JavaScript.

3. Project Overview

Mossback Games is a small independent studio that builds short casual browser games. Our games are spread across different file hosts and game jam pages, so players have no single place to find them and we have no home for the brand. We are hiring a freelance web developer to build Mossback Arcade, a free website where visitors can browse our games, open any one of them, and start playing right away with nothing to download or install.

The website is the product. The developer builds five games for the launch, spread across four categories: Arcade, Puzzle, Board, and Action. Five is the launch set, not a cap. The catalog is set up so the studio can add more games from its own side later by editing one data file and dropping in one game file.

The look is a retro pixel arcade. The page background is a dark texture, buttons and cards carry a neon glow, and titles are set in a pixel font. There is no login and no sign up. Everything the visitor does is stored in their own browser: high scores, favorite games, the chosen arcade skin, the sound setting, a local nickname and avatar, earned tokens, and unlocked badges. There is no server and no database. The visual style is original and must not copy any trademarked or copyrighted product. All assets must be royalty free and openly licensed.

4. Technology Stack

- HTML5 for markup, CSS3 for styling, and vanilla JavaScript (ES2015 or later) for all logic.

- All visitor data stored in the browser using localStorage only.
- Google Fonts: Press Start 2P for the logo, titles, and short labels, and Inter for body text.
- An open icon set such as Material Symbols, Feather Icons, or Heroicons.
- No front end framework such as React, Vue, Angular, or Svelte.
- No UI or CSS framework such as Bootstrap or Tailwind.
- No game engine or game library such as Phaser, Pixi.js, or Kaboom.
- No backend, no server code, and no database.

5. Brand and Visual Style

The site name shown everywhere is Mossback Arcade, written as an original pixel style wordmark next to a small arcade cabinet icon, shown in the banner at the top of this brief. The studio name shown in the footer is Mossback Games.

- The overall look is a retro pixel arcade.
- The page background is a dark texture, such as a carbon fiber or woven pattern, not a flat color.
- Buttons, cards, and panels have a neon outline that glows, in white or in an accent color.
- Headings, the logo, and game titles use the Press Start 2P pixel font.
- Body text, descriptions, and labels use Inter, at a base size of 16 px and a line height near 1.5.
- Each category has its own color, used on its cards and buttons.
- Game cards show a pixel art thumbnail for each game.

Purpose	Name	Hex
Primary neon	Cyan	#36E0E0
Secondary neon	Magenta	#FF3D9A
Token and coins	Gold	#FFC83D
Skin and logo accent	Purple	#9B5DE5
Arcade category	Green	#2FE6A0
Puzzle category	Orange	#FF9A3D
Board category	Blue	#3D9AFF
Action category	Pink	#FF3D9A
Background base	Near black	#0B0E14
Panel and cards	Slate	#141A24
Text	White	#F4F8FF
Muted text	Gray	#8A93A2

6. Arcade Skins

- Three named skins are chosen from a dropdown in the header: Nebula, Synthwave, and Matrix.
- Nebula is the default. It uses deep space purple and cyan over the dark texture, with a slow starfield particle layer.
- Synthwave uses magenta and orange neon over the dark texture, with a faint sunset grid.
- Matrix uses green neon over near black, with a faint layer of falling dots.
- The header skin switcher shows the name of the current skin.
- The chosen skin is saved in localStorage and applied on every page on the next visit.

- Body text keeps a contrast ratio of at least 4.5 to 1 against its background in every skin.

7. Animated Background

- Every page has an animated particle layer behind the content, drawn with CSS or an HTML5 canvas.
- The animated background never sits on top of buttons or links and never blocks clicks.
- The animated background style matches the active skin, for example a starfield in Nebula.
- When the browser reports a reduced motion preference, the animation stops and shows a still version.

8. Site Structure and Files

Deliver the site in a single folder named mossback-arcade with this layout. All six pages share one stylesheet and one main script. Each game lives in its own file under js/games.

```
mossback-arcade/  
  index.html  
  games.html  
  play.html  
  scores.html  
  profile.html  
  about.html  
  css/styles.css  
  js/main.js  
  js/storage.js  
  js/profile.js  
  js/games/snake.js  
  js/games/memory.js  
  js/games/tilemerge.js  
  js/games/lockout.js  
  js/games/reaction.js  
  assets/img/  
  assets/avatars/  
  assets/icons/  
  assets/sounds/  
  data/games.json  
  README.md
```

9. Game Catalog and Extensibility

- The home and library pages build their game cards from data/games.json, not hard coded on each page.
- data/games.json is an array of game objects.
- Each game object has the keys id, title, category, description, thumbnail, controls, scoreLabel, isNew, and isPopular.
- category is one of Arcade, Puzzle, Board, or Action.
- scoreLabel is the unit shown next to the saved best, such as points, moves, or seconds.
- The catalog lists the five launch games.
- A new game is added by appending one object to data/games.json and adding one matching file under js/games.
- Each game file defines a start function that the play page calls with the play area element and a callback used to report the final score.
- The README includes an Adding a game section that lists these two steps.

10. Header, Navigation, and Footer

- Every page has the same header with the arcade cabinet icon and the Mossback Arcade wordmark that links to the home page.
- The navigation has five pill buttons, each with an icon and a label: Home, Games, Leaderboard, Profile, and About.

- The pill button for the current page has a stronger neon outline than the others.
- The header shows the skin switcher as a dropdown that displays the name of the current skin.
- The header shows a sound on or off toggle.
- The header shows a token counter with a coin icon and the current token balance.
- The header shows the visitor avatar and nickname when a nickname is set, or a Set name link when it is not.
- On screens narrower than 768 px the navigation collapses into a menu button.
- Every page has the same footer showing Mossback Games and a copyright line.

11. Sound, Tokens, and Badges

Sound

- The games play short sound effects such as a pickup sound or a hit sound.
- The header sound toggle turns all game sound on or off.
- When sound is off, no game plays any sound.
- The sound setting is saved in localStorage and read on the next visit.

Tokens

- The visitor earns one token each time a game reaches its end state.
- The visitor earns five extra tokens when a game result beats the saved best for that game.
- The token balance is saved in localStorage and the header counter updates after a game ends.

Badges

- There is a fixed set of six badges that unlock from local play.
- First Play unlocks after the visitor finishes any game once.
- Century unlocks when the Snake score reaches 100 points or more in one game.
- Sharp Memory unlocks when Memory Match is finished in 20 moves or fewer.
- Tile Master unlocks when a 256 tile is reached in Tile Merge.
- Unbeaten unlocks after three wins in a row in Grid Lockout.
- Quick Hands unlocks when Reaction Rush reaches 20 hits or more in one round.
- Unlocked badges are saved in localStorage and shown on the Profile page.

12. Home Page (index.html)

- A featured carousel at the top shows a Game of the Day with a title, a short line of text, and a Play button, chosen from the current date so it is the same for every visitor on a given day.
- The featured carousel shows dot indicators and cycles through a few highlighted games.
- A Popular section shows cards for games flagged isPopular.
- A New Games section shows cards for games flagged isNew, each with a New badge.
- The Popular, New Games, and Recently Played sections each have a View all link to the library.
- A Recently Played row is built from the games the visitor has opened, with a message when none have been played.
- Each Recently Played card shows the game best score.
- A Browse by Category row has one button per category, each showing the category icon, the category name, and the number of games in it.

- Each game card shows the thumbnail, the title, the category, and a favorite toggle.
- Clicking a card outside the favorite toggle opens that game play page.

13. Games Library (games.html)

- A grid shows a card for every game in the catalog.
- A search box filters the visible cards by title as the visitor types, matching without case.
- The search updates the grid without reloading the page.
- Category buttons are shown for All, Arcade, Puzzle, Board, and Action.
- A Favorites filter shows only the games the visitor has marked as favorite.
- The search box and the category filter work together on the same grid.
- When a search or filter returns no games, a No games found message is shown.

14. Play Page (play.html)

- The play page reads the game to load from the URL query string, such as play.html?game=snake.
- When the game value is missing or unknown, a message with a link back to the library is shown.
- The play page shows the game title and a how to play note that lists the controls.
- The selected game runs in a play area.
- A high score panel shows the saved best for that game, or a No score yet message when there is none.
- A favorite toggle for the current game is shown.
- Opening a game adds it to the Recently Played list in localStorage.

15. Games

Each game is an original build of a classic style of play. No game copies the art, logo, sound, or name of any branded product. Each game runs inside the play page play area, shows a game over or win message when it ends, has a restart control, and saves its best result to localStorage.

Snake (Arcade)

- The snake moves on a grid one step at a time.
- Pressing an arrow key changes the snake direction.
- The snake cannot reverse straight back into itself.
- Eating a food item makes the snake grow by one segment.
- Eating a food item increases the score.
- The game ends when the snake hits a wall or its own body.
- The saved best is the highest score in points.

Memory Match (Puzzle)

- A grid of face down cards is shown in matching pairs.
- Clicking a card flips it face up.
- Two flipped cards that match stay face up.
- Two flipped cards that do not match flip back down after a short pause.
- A move counter increases on each pair attempt.
- A timer counts the seconds during the round.
- The game is won when all pairs are matched.
- The saved best is the fastest time in seconds in a won game.

Tile Merge (Puzzle)

- A four by four grid of numbered tiles is shown.
- Pressing an arrow key slides all tiles in that direction.
- Two tiles with the same number merge into one tile with the sum.
- A new tile appears after each move that changes the grid.
- The score increases by the value of each merge.
- The game ends when no move can change the grid.
- The saved best is the highest score in points.

Grid Lockout (Board)

- A three by three board is shown and the player marks are X.
- Clicking an empty cell places the player mark.
- A computer opponent places its mark after the player move.
- A win is detected along a row, a column, or a diagonal.
- A draw is detected when the board fills with no winner.
- The saved best is the longest run of wins in a row against the computer.

Reaction Rush (Action)

- A target appears at a random spot in the play area.
- Clicking or tapping the target before it moves scores a hit.
- The score increases by one for each hit.
- A round lasts 30 seconds.
- The final score is shown when the round ends.
- The saved best is the highest number of hits in one round.

16. Leaderboard Page (scores.html)

- The Leaderboard lists every game in the catalog with the best saved result on the device.
- Each saved best is shown with its unit from the scoreLabel field.
- A game with no saved best shows a No score yet line.
- The Leaderboard shows the visitor nickname when one is set.
- The numbers on the Leaderboard match the saved best shown on each game play page.

17. Profile Page (profile.html)

- The visitor can type and save a nickname, which then shows in the header on every page.
- The visitor can pick an avatar from a set of preset character images, which then shows in the header.
- The page shows the current token balance.
- The page shows the six badges and marks which ones are unlocked.
- The page lists the visitor favorite games.
- A reset button clears the visitor saved data after a confirm step.

18. About Page (about.html)

- A short description of Mossback Games is shown.
- One contact line is shown.

19. First Visit Defaults

- On a first visit with empty storage the site uses the Nebula skin and the sound is on.
- On a first visit the token balance is zero, with no badges, no favorites, no recently played games, and no saved scores.
- The site loads without errors when localStorage is empty.

20. Responsive Layout

- The layout works at a mobile width of 320 px and up, a tablet width of 768 px and up, and a desktop width of 1024 px and up.
- Content is centered with a maximum width near 1200 px on large screens.
- Game card grids show one column on a narrow screen and more columns on a wide screen.
- Each game play area fits inside a small screen without overflowing the page.

21. Royalty Free Resource Requirements

All assets must be royalty free and openly licensed under CC0, CC BY, public domain, or a stated free commercial license. The developer must not use any asset from a copyrighted product, a paid asset store, or an unlicensed source. Every external asset must be listed in the README with its name, source link, and license type.

Asset Type	Approved Free Sources	What to Avoid
Fonts	Google Fonts (Press Start 2P, Inter)	Paid fonts, or any font without a free commercial license
Icons	Material Symbols, Feather Icons, Heroicons	Paid icon packs, or icons copied from a branded product
Images, thumbnails, avatars	Original pixel art by the developer, or CC0 images from Pixabay, Unsplash, OpenGameArt, or the Internet Archive	Assets from Pixabay, Unsplash, OpenGameArt, or the Internet Archive that are copyrighted or trademarked
Sound effects	Freesound.org (CC0 or CC BY), Mixkit, ZapSplat	Sounds taken from commercial games or any unlicensed source
Logo and wordmark	Original pixel lettering. The name Mossback Arcade must be used.	Any wordmark that resembles an existing brand mark

22. Required Deliverables

The developer must deliver all of the following items. No partial deliveries are accepted.

#	Deliverable	Description
1	Website source folder	mossback-arcade/ with the six pages, the css, js, assets, and data folders, and a README
2	Six pages	index.html, games.html, play.html, scores.html, profile.html, about.html
3	Five launch games	Snake, Memory Match, Tile Merge, Grid Lockout, Reaction Rush

#	Deliverable	Description
4	Game catalog	data/games.json that drives the cards and supports adding more games
5	Arcade skins	Nebula, Synthwave, and Matrix skins from a header dropdown, saved in the browser
6	Pixel arcade look	Dark textured background, neon outlined buttons and cards, pixel font titles, and an animated particle
7	Local profile	Nickname, preset character avatar, tokens, and badges saved in the browser
8	Persistence	High scores, favorites, skin, sound, nickname, avatar, tokens, and badges saved with localStorage
9	Responsive layout	Works at phone, tablet, and desktop widths
10	README	Run steps, the games and controls, the Adding a game steps, and the asset credit list with licenses
11	Screenshots	At least three screenshots of the finished site
12	Royalty free assets	All fonts, icons, images, and sounds openly licensed with sources listed in the README

23. Out of Scope

The following items are out of scope and must not be included.

- No backend or server code.
- No database.
- No user accounts, login, or sign up.
- No online or networked multiplayer.
- No global online leaderboard. Scores stay local to the device.
- No advertising or analytics scripts.
- No payments or in app purchases.
- No app store packaging such as APK, IPA, or a PWA install flow.
- No front end framework, UI framework, or game engine.
- No content management system or admin panel.

24. Submission Instructions

- Deliver the complete mossback-arcade folder packaged as a single zip archive.
- Include the README with the run steps and the asset credit list.
- Confirm the site has been tested in at least two browsers before delivery.
- Provide at least three screenshots of the finished site.
- All code must be original. Plagiarized or unedited AI boilerplate is not accepted.